CENG381 Stochastic Processes

Finite Markov Chains — Practice 1 (Answer Key)

Q1 Solution

a) Each row must sum to 1:

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Row S:  ? = 1 − 0.3 − 0.2 = 0.5
Row C:  ? = 1 − 0.4 − 0.3 = 0.3
Row R:  ? = 1 − 0.1 − 0.5 = 0.4
```

```
        S      C      R
 S  [ 0.5    0.3    0.2 ]
 C  [ 0.4    0.3    0.3 ]
 R  [ 0.1    0.5    0.4 ]
```

b) $P^{(2)} = P \cdot P$ (entry $p_{ij}^{(2)} = \Sigma_k P_{ik} \cdot P_{kj}$):

Row S of $P^{(2)}$:

$p_{SS}^{(2)} = 0.5\times0.5 + 0.3\times0.4 + 0.2\times0.1 = 0.25+0.12+0.02 = 0.39$

$p_{SC}^{(2)} = 0.5\times0.3 + 0.3\times0.3 + 0.2\times0.5 = 0.15+0.09+0.10 = 0.34$

$p_{SR}^{(2)} = 0.5\times0.2 + 0.3\times0.3 + 0.2\times0.4 = 0.10+0.09+0.08 = 0.27$

Row C of $P^{(2)}$:

$p_{CS}^{(2)} = 0.4\times0.5 + 0.3\times0.4 + 0.3\times0.1 = 0.20+0.12+0.03 = 0.35$

$p_{CC}^{(2)} = 0.4\times0.3 + 0.3\times0.3 + 0.3\times0.5 = 0.12+0.09+0.15 = 0.36$

$p_{CR}^{(2)} = 0.4\times0.2 + 0.3\times0.3 + 0.3\times0.4 = 0.08+0.09+0.12 = 0.29$

Row R of $P^{(2)}$:

$p_{RS}^{(2)} = 0.1\times0.5 + 0.5\times0.4 + 0.4\times0.1 = 0.05+0.20+0.04 = 0.29$

$p_{RC}^{(2)} = 0.1\times0.3 + 0.5\times0.3 + 0.4\times0.5 = 0.03+0.15+0.20 = 0.38$

$p_{RR}^{(2)} = 0.1\times0.2 + 0.5\times0.3 + 0.4\times0.4 = 0.02+0.15+0.16 = 0.33$

```
P⁽²⁾ =
  [ 0.39   0.34   0.27 ]   (row sums: 1.00 ✓)
  [ 0.35   0.36   0.29 ]   (row sums: 1.00 ✓)
  [ 0.29   0.38   0.33 ]   (row sums: 1.00 ✓)
```

c) $p_{SR}^{(3)}$ via Chapman-Kolmogorov (n=2, m=1):

```
p_{SR}⁽³⁾ = p_{SS}⁽²⁾·P_{SR} + p_{SC}⁽²⁾·P_{CR} + p_{SR}⁽²⁾·P_{RR}
          = 0.39×0.2 + 0.34×0.3 + 0.27×0.4
```

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        = 0.078 + 0.102 + 0.108
        = 0.288
```

Q2 Solution

a) Transition diagram (description):

States 1 and 2 each have self-loops (prob ½) and arrows to each other (prob ½). State 3 has arrows to states 2, 3, 4 (each prob ⅓). States 4 and 5 communicate with each other: 4→5 (prob ⅔), 5→4 (prob ⅔), 4→4 (prob ⅓), 5→5 (prob ⅓). There are no arrows from {1,2} to {3,4,5} or from {4,5} back to {1,2,3}.

b) Communicating classes:

Class {1, 2}: $P(1,2)=½>0$ and $P(2,1)=½>0$, so 1↔2. No transitions leave this class ($P(1,3)=P(2,3)=\ldots=0$). → CLOSED → Positive Recurrent.

Class {3}: From 3 we can reach 2 (prob ⅓) or 4 (prob ⅓). Neither 2 nor 4 can return to 3. State 3 can never return to itself with prob 1. → Transient.

Class {4, 5}: $P(4,5)=⅔>0$ and $P(5,4)=⅔>0$, so 4↔5. No transitions leave this class. → CLOSED → Positive Recurrent.

c) Limiting distributions:

Within class {1, 2} (restricted matrix, solve $\pi \cdot P = \pi$):

$\pi_1 = ½\pi_1 + ½\pi_2$ and $\pi_1+\pi_2=1$ → $\pi_1=\pi_2$ → $\pi_1=\pi_2=½$

Within class {4, 5}:

$\pi_4 = ⅓\pi_4 + ⅔\pi_5$ → $⅔\pi_4=⅔\pi_5$ → $\pi_4=\pi_5=½$

Absorption from state 3 (let h = P(absorbed by {1,2} | start 3)):

$h = ⅓\cdot 1 + ⅓\cdot h + ⅓\cdot 0$ → $⅔h = ⅓$ → $h = ½$

So from state 3: absorbed into {1,2} with prob ½, into {4,5} with prob ½.

Long-run probabilities starting from state 3:

$\pi_1=\pi_2=¼$, $\pi_3=0$, $\pi_4=\pi_5=¼$ (weighted average of the two class limits).

Q3 Solution

a) Transition matrix P (order G, N, B):

$$
\begin{array}{c|ccc}
 & G & N & B \\
\hline
G & 0.6 & 0.3 & 0.1 \\
N & 0.4 & 0.3 & 0.3 \\
B & 0.2 & 0.5 & 0.3
\end{array}
$$

b) Two-step distribution from N (Monday → Wednesday):

$\mu_0 = (0, 1, 0)$ (start in N on Monday)

$\mu_1 = \mu_0 \cdot P = (0\cdot0.6+1\cdot0.4+0\cdot0.2,\ 0\cdot0.3+1\cdot0.3+0\cdot0.5,\ 0\cdot0.1+1\cdot0.3+0\cdot0.3)$

$= (0.4,\ 0.3,\ 0.3)$ [Tuesday distribution]

$$
\begin{aligned}
\mu2[G] &= 0.4\times0.6 + 0.3\times0.4 + 0.3\times0.2 = 0.24+0.12+0.06 = 0.42 \\
\mu2[N] &= 0.4\times0.3 + 0.3\times0.3 + 0.3\times0.5 = 0.12+0.09+0.15 = 0.36 \\
\mu2[B] &= 0.4\times0.1 + 0.3\times0.3 + 0.3\times0.3 = 0.04+0.09+0.09 = 0.22 \\
\text{Check: } & 0.42+0.36+0.22 = 1.00 \quad \checkmark
\end{aligned}
$$

Wednesday distribution: P(G)=0.42, P(N)=0.36, P(B)=0.22.

c) $p_{GB}^{(3)}$ via Chapman-Kolmogorov:

First, compute row G of $P^{(2)}$:

$$
\begin{aligned}
p_{GG}^{(2)} &= 0.6\times0.6+0.3\times0.4+0.1\times0.2 = 0.36+0.12+0.02 = 0.50 \\
p_{GN}^{(2)} &= 0.6\times0.3+0.3\times0.3+0.1\times0.5 = 0.18+0.09+0.05 = 0.32 \\
p_{GB}^{(2)} &= 0.6\times0.1+0.3\times0.3+0.1\times0.3 = 0.06+0.09+0.03 = 0.18 \\
\text{Check: } & 0.50+0.32+0.18 = 1.00 \quad \checkmark
\end{aligned}
$$

$$
\begin{aligned}
p_{GB}^{(3)} &= p_{GG}^{(2)}\times P_{GB} + p_{GN}^{(2)}\times P_{NB} + p_{GB}^{(2)}\times P_{BB} \\
&= 0.50\times0.1 + 0.32\times0.3 + 0.18\times0.3 \\
&= 0.050 + 0.096 + 0.054 \\
&= 0.200
\end{aligned}
$$